

Wei Li

Professor / PhD Supervisor / Nanjing University of Posts and Telecommunications (NJUPT)

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Summary

National High-level Young Talent (Overseas), Distinguished Professor of Jiangsu Province, China.
 Director of the Smart Mechatronics & Integration Lab.
 IEEE Senior Member
 Expertise: Mechatronics, robotics, MEMS, and flexible electronics.
 Former postdoc at UC Berkeley & LBNL, and Tenure-Track Assistant Professor at the University of Vermont.

Education

PhD	Michigan State University (MSU) , Mechanical Engineering • Advisor: Prof. Nelson Sepulveda	Michigan, USA 2014 – 2017
PhD	Huazhong University of Science and Technology (HUST) , Mechatronic Engineering • Advisor: Prof. Xuedong Chen (Member of the Chinese Academy of Engineering)	Wuhan, China 2009 – 2013
B.S.	Huazhong University of Science and Technology (HUST) , Mechanical Engineering	Wuhan, China 2004 – 2008

Experience

Nanjing University of Posts and Telecommunications (NJUPT) , Professor, PhD Supervisor • College of Integrated Circuit Science and Engineering • Role: Director of Intelligent Mechatronic Systems Laboratory • Honors: National High-level Young Talent, Distinguished Professor of Jiangsu Province	Nanjing, China 2022 – present 4 years
Huazhong University of Science and Technology (HUST) , Adjunct Professor • School of Mechanical Science and Engineering • State Key Laboratory of Intelligent Manufacturing Equipment and Technology	Wuhan, China 2025 – present 1 year
State Key Laboratory of Flexible Electronics (LoFE) , Professor	Nanjing, China 2025 – present 1 year
University of Vermont (UVM) , Tenure-Track Assistant Professor, • Dept. of Mechanical Engineering	Vermont, USA 2020 – 2022 2 years
University of California, Berkeley (UC Berkeley) , Postdoctoral Scholar • Advisors: Prof. Kristofer Pister & Prof. Michel Maharbiz • Dept. of Electrical Engineering and Computer Sciences (EECS) • Berkeley Sensor and Actuator Center (BSAC)	California, USA 2017 – 2020 3 years
Lawrence Berkeley National Laboratory (LBNL) , Postdoctoral Scholar • Advisor: Dr. Paul Ashby • Molecular Foundry	California, USA 2018 – 2019 1 year

Funded Projects

National Science Fund for Excellent Young Scholars (Overseas) PI, National Natural Science Foundation of China (NSFC), NJUPT, Funding: 2,000K RMB	2022 – 2025
Advanced Wireless - Integration with Infrastructure System Co-PI, NSF (2119485), UVM, Funding: 3,995K USD	2021 – 2025
Jiangsu Provincial Dual-Innovation Team Co-PI, Jiangsu Province (JSSCTD202351), NJUPT, Funding: 3,000K RMB	2023 – 2026
Intelligent Robot Skin Integrated with Triple Sensing Modes of Heat, Force and Wave PI, Research Startup Fund for High-level Talents (NY223124), NJUPT, Funding: 2,000K RMB	2024 – 2027
Distinguished Professor of Jiangsu Province Program PI, Jiangsu Province (RK118STP22006), Funding: 1,000K RMB	2022 – 2025
High-Robust Flexible Intelligent Perception System for Low-Altitude Complex Environments Co-PI, State Key Laboratory of Flexible Electronics (LoFE), Funding: 2,000K RMB	2025 – 2029
Short-Range Independent Microrobotic Platform Participant, DARPA (HR001118S0048), UC Berkeley, Funding: 3,200K USD	2019 – 2020
Mobility and Power for Micro Robots Participant, U.S. Army Research Laboratory (PN72880-SE-ARL), UC Berkeley, Funding: 371K USD	2018 – 2019
Multiscale Self-Assembly for Strong and Tough Living Materials Participant, DARPA (FP00004778), UC Berkeley, Funding: 3,380K USD	2017 – 2018
Development of Intrinsically Stretchable, Active-Matrix OLED Displays Participant, NSF (1549888), MSU, Funding: 100K USD	2015 – 2016
Enabling a New Technology for Reconfigurable RF Front-Ends and Antenna Array Systems Through Phase-Change Materials Participant, NSF (1310257), MSU, Funding: 200K USD	2014 – 2017
Development of a New Class of Thermal Transducers Through Phase-Change Materials Participant, NSF (1306311), MSU, Funding: 360K USD	2014 – 2017
Novel Vanadium Dioxide-based Self-Sensing Microactuators: Modeling, Control, and Application to Micromanipulation Participant, NSF (1301243), HUST, Funding: 3,000K RMB	2014 – 2017
Fundamental Research on High Performance Aerostatic Bearings System Participant, State Key Program of National Natural Science of China (51235005), HUST, Funding: 4,200K RMB	2013
Fundamental Research on High Performance Digital Manufacturing Equipment Participant, Science Fund for Creative Research Groups of the NSFC (51121002), HUST, Funding: 8,500K RMB	2011 – 2013
Modeling and Control of Multi-body and Multi-field System with Large Inertia Participant, National Basic Research Program of China (973 Program) (2009CB724205), HUST, Funding: 3,800K RMB	2009 – 2013

Structural Analysis and Optimization of Dual Wafer Stages	2009 – 2013
Participant, National Science and Technology Major Project of China (2009ZX02208-007), HUST, Funding: 15,100K RMB	
High Performance Ultra-precision Vibration Isolation System	2009 – 2013
Participant, National Science and Technology Major Project of China (2009ZX02208-010), HUST, Funding: 22,100K RMB	

Publications

Enabling the 6G and IoT-Verse: Non-Radiative Dielectric (NRD) Waveguides for Millimeter-Wave Communications	2026
T. Bashir, W. Li , I. Patwary, A. Iguchi, Y. Tsuji, T. Kashiwa 10.1109/JIOT.2026.3676274 (IEEE Internet of Things Journal)	
Ultra-Miniaturized Dual-Band MIMO Antenna for Biomedical Implantable Devices in Wireless Health Monitoring Systems	2026
T. Bashir, S. Chen, G. Feng, Y. Cao, W. Li 10.3390/bios16030163 (Biosensors, Vol. 16)	
A Highly Compact and Isolated Triple-Band MIMO Antenna for Wireless Capsule Endoscopy and Cardiac Implant	2026
T. Bashir, G. Feng, S. Chen, Y. Cao, W. Li 10.3390/mi17030296 (Micromachines, Vol. 17)	
Embedded-AI-Driven On-Site Traumatic Brain Injury Assessment Using Wireless Flexible Wearable Sensors for Real-Time Impact Force and Acceleration Estimation	2026
X. Chen, S. Wang, Y. Zhang, Y. Ye, J. Ye, W. Li , et al. 10.1109/jiot.2026.3655010 (IEEE Internet of Things Journal, Vol. 13)	
An Ion Discharge-Driven Thruster Based on a Lithium Niobate Piezoelectric Transformer	2025
Q. Tao, X. Wang, Y. Gu, W. Li 10.3390/mi16030277 (Micromachines, 16(3))	
High-thrust ultralight ion wind engine toward controllable aerial microrobots	2025
Q. Tao, Z. Chen, Y. Cao, W. Li 10.1016/j.energy.2024.139027 (Energy, Vol. 339)	
Implantable MIMO Antenna for WMTS and ISM Bands in Leadless Cardiac Pacemaker System	2025
C. Shunbiao, T. Bashir, W. Li 10.1109/ucmmt67044.2025.11287781 (2025 18th IEEE UCMMT)	
A High-Performance Miniaturized Frequency Shift Detection System for QCM-Based Gravimetric Sensing	2025
C. Gao, S. Fan, W. Li , et al. 10.1002/adsr.202400148 (Advanced Sensor Research, Vol. 4)	
A Multi-Functional VOC Sensor Based on Cascaded Quartz Crystal Resonators	2025
M. Fu, Z. Ma, C. Gao, Y. Ye, W. Li , et al. 10.1109/LED.2025.3528024 (IEEE Electron Device Letters)	
Highly Stretchable Shape Memory Hydrogels with Silk-Fibroin Semi-Interpenetrating Networks for Drug-Delivery Microrobot	2025
Z. Feng, J. Xie, W. Li 10.1109/ucmmt67044.2025.11287593 (2025 18th IEEE UCMMT)	

Silk Fibroin-Regulated Shape Memory Composite Hydrogels with Tunable Recovery Time Z. Feng, W. Huang, F. Lv, W. Li 10.1002/adfm.202531055 (Advanced Functional Materials)	2025
Miniaturized Implantable Mimo Antenna for Gastrointestinal Wireless Capsule Endoscopy G. Feng, T. Bashir, W. Li 10.1109/ucmmt67044.2025.11287480 (2025 18th IEEE UCMMT)	2025
A High-Power-Density Ultralow-Frequency Energy Harvester Based on a Magnetic Rotor With Built-In Eccentricity and Ferromagnetic Fillers S. Fan, H. Chi, Y. Yao, W. Li , et al. 10.1109/tmech.2025.3618933 (IEEE/ASME Transactions on Mechatronics)	2025
Room-Temperature Single-Molecule Mass Detection via Feedback Control of ZnO Nanowire Resonator X. Cai, X. Wang, Z. Feng, Y. Cao, X. Zheng, W. Li 10.1021/acsami.5c06688 (ACS Applied Materials & Interfaces)	2025
Exploring the application of ferroelectret nanogenerators in medical engineering X. Cai, X. Han, J. Xie, Y. Cao, W. Li 10.1002/flm2.44 (FlexMat)	2025
Radiation Diversity Enabled Self-Isolated Compact Dual-Band Cubic MIMO Antenna for Wireless Biomedical Implants T. Bashir, W. Li , T. Xia 10.1109/jmw.2025.3583891 (IEEE Journal of Microwaves)	2025
Compact Dual-Band 3D MIMO Antenna for Wireless Biomedical Implantable Devices T. Bashir, G. Feng, C. Shunbiao, W. Li 10.1109/ucmmt67044.2025.11286680 (2025 18th IEEE UCMMT)	2025
Self-powered highly stretchable ferroelectret nanogenerator towards intelligent sports Y. Wang, X. Cai, Y. Guo, W. Li , et al. 10.1016/j.nwnano.2024.100053 (Nano Trends, Vol. 8)	2024
Ultrathin, Stretchable, and Twistable Ferroelectret Nanogenerator for Facial Muscle Detection Z. Song, X. Cai, Z. Chen, Z. Zhu, Y. Cao, W. Li 10.3390/nanoenergyadv4040021 (Nanoenergy Advances, Vol. 4)	2024
Trenched microwave resonator integrated with porous PDMS for detection and classification of VOCs with enhanced performance Z. Ma, M. Fu, C. Gao, W. Li , et al. 10.1016/j.jhazmat.2024.134553 (Journal of Hazardous Materials, Vol. 472)	2024
Microrobotic Flight Enabled by Ultralight Ion Thrusters with High Thrust-to-Weight Ratio and Low Fabrication Cost Y. Gu, X. Cai, K. Thakuri, W. Li , et al. 10.1109/ICRA57147.2024.10611166 (2024 IEEE ICRA)	2024

Evaluating head impact intensities and accelerations using flexible wearable sensors for traumatic brain injury assessment W. Du, S. Wang, H. Chi, W. Li , et al. 10.1016/j.sna.2024.115443 (Sensors and Actuators A: Physical, Vol. 373)	2024
Flexural-Mode Piezoelectric Resonators: Structure, Performance, and Emerging Applications in Physical Sensing Technology X. Cai, Y. Wang, Y. Cao, W. Yang, T. Xia, W. Li 10.3390/s24113625 (Sensors, Vol. 24)	2024
Leveraging Ferroelectret Nanogenerators for Acoustic Applications Z. Song, X. Cai, Y. Wang, W. Yang, W. Li 10.3390/mi14122145 (Micromachines, Vol. 14)	2023
Ferroelectret nanogenerators for the development of bioengineering systems W. Li , Y. Cao, C. Wang, N. Sepúlveda 10.1016/j.xcrp.2023.101388 (Cell Reports Physical Science)	2023
Thin Film Piezoelectric Nanogenerator Based on (100)-Oriented Nanocrystalline AlN Grown by Pulsed Laser Deposition at Room Temperature W. Li , Y. Cao, N. Sepúlveda 10.3390/mi14010099 (Micromachines, Vol. 14)	2023
Recent Advances in the Application of Piezoelectric Materials in Microrobotic Systems A. Fath, T. Xia, W. Li 10.3390/mi13091422 (Micromachines, Vol. 13)	2022
Nanomechanical testing of freestanding polymer films: in situ tensile testing and Tg measurement N.R. Velez, F.I. Allen, W. Li , et al. 10.1557/s43578-021-00163-z (Journal of Materials Research, Vol. 36)	2021
Quick self-assembly of bio-inspired multi-dimensional well-ordered structures induced by ultrasonic wave energy C. Murphy, Y. Cao, N. Sepúlveda, W. Li 10.1371/journal.pone.0246453 (PLoS ONE, Vol. 16)	2021
15 Millinewton Force, 1 Millimeter Displacement, Low-Power MEMS Gripper C.B. Schindler, H.C. Gomez, D. Teal, W. Li , K.S.J. Pister 10.1109/MEMS46641.2020.9056128 (2020 IEEE MEMS)	2020
Performance of Self-Powered, Water-Resistant Bending Sensor Using Transverse Piezoelectric Effect of Polypropylene Ferroelectret Polymer Y. Cao, W. Li , N. Sepulveda 10.1109/JSEN.2019.2933174 (IEEE Sens. J., Vol. 19)	2019
Flexible Ferroelectret Polymer for Self-Powering Devices and Energy Storage Systems Y. Cao, J. Figueroa, W. Li , et al. 10.1021/acsami.9b02233 (ACS Applied Materials & Interfaces)	2019
Understanding the dynamic response in ferroelectret nanogenerators to enable self-powered tactile systems and human-controlled micro-robots Y. Cao, J. Figueroa, W. Li , et al. 10.1016/j.nanoen.2019.06.048 (Nano Energy, Vol. 63)	2019

Impact-activated programming of electro-mechanical resonators through ferroelectret nanogenerator (FENG) and vanadium dioxide Y. Cao, W. Li , et al. 10.1016/j.nanoen.2017.10.066 (Nano Energy, Vol. 43)	2018
Fully Printed Silver-Nanoparticle-Based Strain Gauges with Record High Sensitivity S. Zhang, L. Cai, W. Li , et al. 10.1002/aelm.201700067 (Advanced Electronic Materials, Vol. 3)	2017
Nanogenerator-based dual-functional and self-powered thin patch loudspeaker or microphone for flexible electronics W. Li , D. Torres, R. Diaz, Z. Wang, C. Wu, C. Wang, Z.L. Wang, N. Sepulveda 10.1038/ncomms15310 (Nature Communications, Vol. 8)	2017
Flexible and biocompatible polypropylene ferroelectret nanogenerator (FENG): On the path toward wearable devices powered by human motion W. Li , D. Torres, T. Wang, C. Wang, N. Sepulveda 10.1016/j.nanoen.2016.10.007 (Nano Energy, Vol. 30)	2016
A Monolithic Self-Sensing Precision Stage: Design, Modeling, Calibration, and Hysteresis Compensation X. Chen, W. Li 10.1109/TMECH.2014.2306231 (IEEE/ASME Transactions on Mechatronics, Vol. 20)	2015
Inverse compensation for hysteresis in piezoelectric actuator using an asymmetric rate-dependent model W. Li , X. Chen, Z. Li 10.1063/1.4833399 (Review of Scientific Instruments, Vol. 84)	2013
Compensation of hysteresis in piezoelectric actuators without dynamics modeling W. Li , X. Chen 10.1016/j.sna.2013.04.036 (Sensors and Actuators A: Physical, Vol. 199)	2013

Patents

Flexible Stretchable Ferroelectret and its Preparation Method (CN118870275B) W. Li , Z. Song, X. Cai, Y. Wang	Oct 2025
Flexible Stretchable Ferroelectret Multi-axis Force Sensor (CN120651412A) W. Li , Z. Song, J. Xie	Sept 2025
Smart Ping Pong Paddle Prepared Based on Ferroelectret Flexible Material (CN120532096A) W. Li , Y. Wang, J. Xie	Aug 2025
Swimming Robot Based on Dielectric Barrier Discharge Propulsion (CN120246213A) W. Li , Q. Tao, Y. Song, X. Cai, Y. Gu, Z. Chen	July 2025
Origami-Based Ion Wind Aircraft (CN120024530A) W. Li , X. Cai, Y. Shen, M. Tian, R. Xu	May 2025
Tunable Vanadium Dioxide Nanowire Sensor (CN119959308A) X. Cai, W. Li , Y. Shen	May 2025
Electromagnetically Excited ZnO Nanowire Resonator (CN119675620A) X. Cai, W. Li	Mar 2025

Highly Stretchable Self-Powered Muscle Patch for Human Motion Monitoring (CN119279568A) W. Li, Y. Wang, Y. Song, J. Xie, X. Cai, Y. Feng, X. Han, Z. Song	Jan 2025
Tensile Mechanical Testing Device and Operation Method (CN119133003A) W. Li, X. Cai, Z. Cao	Dec 2024
Flexible Stretchable Ferroelectret and its Preparation Method (CN118870275A) W. Li, Z. Song, X. Cai, Y. Wang	Oct 2024
Bionic Starfish Robot (CN118238915A) W. Li, Z. Chen, Y. Gu, X. Cai	June 2024
Multi-legged Bionic Robot and its Manufacturing and Control Method (CN117066295A) W. Li, Y. Wang, Y. Guo, Y. Gu, Z. Song	Nov 2023
Ultrasonic Sensing Flexible Electronic Skin and Robot (CN116430392A) W. Li, Y. Wang	July 2023
Multi-stage Multi-Degree-of-Freedom Ion Wind Aircraft (CN116215904A) W. Li, Y. Gu	June 2023
Vibration Isolator Containing Magnetic Levitation Unit (CN201210298969.2) X. Chen, W. Li, L. Zeng, X. Li, Z. Xu, X. Zhao, J. Wang	Jan 2013
Ultra-Precision Stage (CN201110090892.5) X. Chen, W. Li, X. Luo, X. Li	Sept 2011
Precision Active Vibration Damping Device (CN201110058574.0) X. Chen, W. Li, Z. Xu, Z. Yuan, W. Wu	Aug 2011
Ultra-Precision Stage with Air Floating Support and Electromagnetic Drive (CN201120451882.5) X. Chen, W. Li, X. Luo, X. Li	July 2012
Dual Workpiece Stage Clamping Mechanism (CN201010575399.8) Z. Yuan, W. Li, X. Wu	June 2012
Vibration Damping Device and Lithography Apparatus Using Same (CN201010102421.7) F. Liao, X. Li, W. Li, L. Liu	Aug 2011
Vibration Damper for Vacuum Environment (CN200910273182.9) X. Li, F. Liao, X. Chen, W. Li, L. Liu, X. Luo, Z. Xu, L. Zeng	June 2010
Horizontal 2-DOF Vibration Isolation Mechanism (CN200910273183.3) X. Chen, X. Li, L. Liu, W. Li, F. Liao, X. Luo, L. Zeng, Z. Xu	May 2010

Teaching

- Microelectromechanical Systems (MEMS)
- Introduction to Robotics
- Control Systems
- Micro/Nano Robotics
- Integrated Circuit Fabrication Process

Service

- Guest Editor, Micromachines, Special Issue ‘Piezoelectric MEMS/NEMS — Materials, Devices, and Applications’ (2022-2026)
- Guest Editor, Frontiers in Bioengineering and Biotechnology, Special Issue ‘Nanogenerator Technologies bridging Bioengineering and Human-Machine Integration’ (2025)
- Section Chair, International Conference on Integrated Circuits and Microsystems (IEEE ICICM 2023)
- Local Chair, International Symposium on Automation, Mechanical and Design Engineering (SAMDE 2023)
- Reviewer of IEEE T-MECH, IEEE T-RO, IEEE-ICRA, Adv. Funct. Materials, APL, etc.
- Mentor of Senior Experience in Engineering Design (SEED), UVM
- Member of IEEE Robotics and Automation Society
- Member of IEEE Industrial Electronics Society
- Member of Dielectrics and Electrical Insulation Society
- Member of IEEE Antennas and Propagation Society
- Member of IEEE Microwave Theory and Technology Society
- Young Editorial Board Member of Nano-Micro Letters, FlexMat, Soft Science, etc.

Media Coverage

- Xinhua Net
- People’s Daily (Overseas Edition)
- The Wall Street Journal
- The Christian Science Monitor
- Voice of America
- EENews Europe
- National Science Foundation (NSF)
- American Association for the Advancement of Science (AAAS)
- Daily Mail (UK)
- Phys.org
- ScienceDaily
- National Public Radio (NPR)